

Failure Modes in Untargeted MS1 Feature-Detection Algorithms

Alex Solivaïs

July 6, 2026

Abstract

This chapter is an empirical study of *why* an untargeted (de-novo, no-identification) MS1 feature-detection pipeline fails to recover peptide features, and of what fixes each failure demands. The vehicle is a pure-Rust reimplementation of FlashLFQ extended with a three-stage untargeted pipeline (detect $\rightarrow$ refine $\rightarrow$ resolve), validated by matching its output against FlashLFQ's identification-driven quantified peaks on a carbonic-anhydrase tryptic HCD run acquired on a Thermo Orbitrap Fusion Lumos. Over one extended investigation, reference recall rose from 589/620 (95.0%) to 602/620 (97.1%) and charge-matched recall from 564 (91.9%) to 584 (94.2%) by diagnosing and repairing a sequence of distinct failure modes. Rather than present only the working pipeline, this chapter organises those failures into a taxonomy by pipeline stage and by root mechanism, and for each gives symptom, mechanism, concrete evidence (named peptide with masses, m/z , retention time, and fit scores), the fix, and the generalisable lesson. It devotes a dedicated section to *negative* results — fixes that regressed recall and were therefore reverted — because those are as instructive as the successes and prevent repeated effort. It closes with design principles for isotope-envelope feature detectors and a candid accounting of the misses that remain. Where a claim rests on a single traced example, this is stated explicitly.

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1 Scope, pipeline, and validation methodology

1.1 The pipeline under study

The system is a de-novo MS1 feature detector: it receives centroided MS1 scans and must produce a list of peptide-level features (monoisotopic neutral mass, charge state(s), retention-time (RT) extent, intensity) with *no* peptide identifications as input. It has three stages.

1. **Detect** (`trace_kernel.rs`). A sparse matched-filter / trace-kernel detector. Every centroid is a candidate seed; seeds are processed tallest-first. For each seed, a charge is decided by scoring an isotope-comb $\times$ RT-Gaussian hypothesis over $z=z$ for $z \in [z_{\min}, z_{\max}]$ and keeping the highest response (*cross-charge non-maximum suppression*, cross- z NMS). The winning feature’s peaks are then *claimed* — removed from further seeding — so overlapping harmonics cannot re-fire. The default scorer is `RawSum`, an un-normalised $\sum_k \sum_s w_k g_s I_{k,s}$ over comb teeth k and RT scans s . A coverage-target stop and a chromatographic-persistence gate (≥ 2 distinct scans) bound the feature population.
2. **Refine** (`feature_refinement.rs`, `isotope_shift_decon.rs`). Each detected feature is deconvolved on an averaged apex composite (≤ 3 scans, apex ± 1) and/or the single apex scan. A FlashLFQ-style ^{13}C isotope-shift search ($-1/0/+1$, widened to $-2 \dots +2$ at high charge) places the monoisotope; an *envelope-fit cosine* metric selects charge and monoisotope. A high-charge *mono walk-back* corrects heavy-peptide off-by-one-too-high placements.
3. **Resolve** (`feature_refinement::resolve_charge_state_consensus`). Refined features are knitted into peptide-level features by single-linkage grouping on co-elution plus off-by-one-aware mass agreement, then a cross-charge mass vote settles the consensus monoisotopic mass.

1.2 Validation data and metric

The benchmark is a carbonic-anhydrase (“CA”) tryptic digest, HCD fragmentation, 10-minute gradient, acquired on a Thermo Lumos (2689 MS1 scans, uncalibrated `.raw`). The reference is

FlashLFQ’s `AllQuantifiedPeaks.tsv` from a wide-tolerance MetaMorpheus/GPTMD search (634 rows; ~ 620 quantifiable peaks after filtering MBR/decoy rows with non-numeric RT/intensity). **Recall** is the fraction of reference peaks rediscovered by the untargeted output within ± 20 ppm mass and ± 0.3 min RT; **charge-matched recall** additionally requires the charge state to agree.

Two methodological cautions are load-bearing throughout this chapter.

- **A systematic $\sim +10$ ppm calibration offset.** The reference **Peptide Monoisotopic Mass** is the *theoretical* peptide mass; the raw file is *uncalibrated*. Untargeted masses therefore run systematically $\sim +10$ ppm high against the reference. This is instrument calibration, not a detector error. Concretely, EFLDVIK’s theoretical reference mass is 862.480 Da while the untargeted pipeline resolves it at 862.488 Da — a ~ 9 ppm offset, not a mono-placement error. Matches are accepted within an integer- ^{13}C offset plus residual ppm.
- **Compare on observed m/z , not theoretical mass.** The reference **Peak MZ** is the *most-abundant* isotope m/z , not the monoisotopic m/z ; for $z \geq 2$ it sits ~ 1 ^{13}C above the mono ($\approx +0.5$ Th at $z=2$). Observed-vs-observed m/z agreement is dead-on (± 1 – 2 ppm); the residual $\sim +10$ ppm appears only in theoretical-vs-observed comparisons.

1.3 How to read this chapter

The failure modes are organised by pipeline stage (Sections 2–7), because the stage locates the fix, but the deeper organising axis is *root mechanism*: most failures reduce to one of four mechanisms — (i) a scoring window that cannot see signal on one side of the hypothesis; (ii) harmonic ($z \leftrightarrow 2z$) ambiguity inherent to isotope-comb matching; (iii) anchoring a deconvolution on the wrong (stronger, foreign) peak; and (iv) aggregating cross-charge evidence by quantity or intensity rather than by fit quality. Table 1 maps every documented failure to its stage, mechanism, traced peptide, and fix; Table 2 records the recall progression across the fixes.

2 Monoisotope placement

2.1 Heavy-peptide off-by-one-too-high, and the window blind spot

Symptom. Heavy, highly-charged peptides are recovered at the correct charge and approximate mass but with the monoisotope placed one or two ^{13}C units too high, so the neutral mass is ~ 1 – 2 Da heavy and the reference match fails.

Mechanism. For a heavy peptide the isotopic envelope *mode* (its tallest tooth) sits one or more ^{13}C above the true monoisotope. The detector seeds on the tallest peak; the shift search, when its anchor or slice differs from the ideal, can then place the mono on a peak one or two ^{13}C above the true one. Crucially, the envelope-fit cosine metric that scores placements uses a window that begins at $\text{mono} - 0.5 \cdot \Delta$ (where Δ is the isotope spacing). A too-high placement’s window therefore *never sees* the strong true-mono peak sitting just below it: the unexplained signal beneath the hypothesis is off-grid, so the wrong placement scores as cleanly as the right one. This is the *window blind spot*: a fit or correlation window that does not extend below the monoisotope is structurally incapable of detecting an off-by-one-too-high.

Evidence. Traced on **ECCHGDLLECADDRADLAK** (reference monoisotopic mass 2246.9355 Da, $z=4$, RT 12.153 min; mono m/z 562.741, +1 at 562.997, +2 at 563.243). The detector seeded the

mono on the +2 isotope (562.247, mass ≈ 2248.96). A placement one ^{13}C too high (mass 2247.9578) scores cosine 0.934 — an apparently acceptable fit — precisely because the strong 562.746 peak ($\approx 8.0 \times 10^6$ intensity) lies just below the window and is invisible to the metric.

Fix and why it works. `walkback_mono_high_charge` re-scores the chosen mono against its 1- and 2- ^{13}C -lower alternatives and adopts the *lowest-mass* hypothesis whose fit is within a small margin (`WALKBACK_FIT_MARGIN` = 0.01) of the best. The lower alternatives’ windows *do* include the peaks beneath the original placement, so they can finally see the unexplained signal that condemned it. The check is self-protecting: a correctly-placed mono’s lower alternatives predict a leading tooth where nothing is observed, so they fit far worse and a good mono is never walked back. It is gated on mass ≥ 2200 Da and $|z| \geq 3$ — the heavy-peptide regime where the mode sits well above the mono. On ECC the walk-back recovers 2246.953 Da (a dead-on reference match within the calibration offset).

Generalisable insight. A model-selection metric must be able to see signal on *both* sides of the hypothesis it scores. A one-sided window silently converts an entire directional error class (here, mono-too-high) into a blind spot; the fix is either to widen the window (but see Section 8.1 for why naïve widening regressed) or to add an explicit, gated re-scoring against lower-mass alternatives whose windows *do* reach the missing evidence.

3 Charge assignment: harmonic ambiguity

Isotope-comb matching has an inherent degeneracy: the teeth of a charge- $z=z$ envelope (spacing $\Delta = 1.00335 \text{ Da}/z$) are a *subset* of the charge- $z=2z$ grid (spacing $\Delta/2$). A $z=z$ hypothesis and a $z=2z$ hypothesis can therefore both “explain” the same peaks. Which way the ambiguity bites depends on what fills — or fails to fill — the intervening teeth.

3.1 Charge-halving (real $z=2$ labelled $z=1$)

Symptom. A genuine light $z=2$ peptide is detected but assigned $z=1$, halving its neutral mass, so it never matches the true-mass reference. This class accounted for ~ 10 peptides (light $z=2$, 788–907 Da, $m/z \sim 395$ –455).

Mechanism. A $z=2$ envelope’s *even* isotopes (m/z 432.25, 433.25, 434.25, ...) themselves form a valid 1.0- Th $z=1$ comb. The `RawSum` charge scorer (Eq. 1) rewards matched intensity but never penalises the strong *intervening* $z=2$ peaks (the +1 at 432.75) that a $z=1$ hypothesis leaves unexplained. $z=2$ explains every observed peak; $z=1$ explains every *other* one and ignores the rest — yet $z=1$ out-scores $z=2$. Refine and resolve then faithfully carry the halved mass.

$$\text{RawSum}(z) = \sum_k \sum_s w_k g_s I_{k,s}, \quad (1)$$

un-normalised and never decremented, is blind to both missing predicted teeth and unexplained observed peaks.

Evidence. Traced on **EFLDVIK** (reference mass 862.480 Da theoretical, true $z=2$, RT 14.70 min; most-abundant m/z 432.251). The strongest feature at that m/z (432.28, intensity 8.7×10^8) was

seeded and assigned $z=1 \rightarrow \text{mono } 431.28 \text{ Da} = \text{half the true } 862.48 \text{ Da}$. Raising the coverage target ($0.90 \rightarrow 0.95$; $96\,000 \rightarrow 143\,000$ features) recovered *zero* additional reference peaks and left the miss set byte-identical — proof that this is a charge-scoring failure, not a detection-depth one. The envelope-fit cosine cleanly ranks the charges: $z=2 \ 0.984 > z=1 \ 0.905 > z=3 \ 0.718$ (see the companion note *An Envelope-Fit Metric for Charge-State and Monoisotope Selection* for the full derivation and the completeness/explained-fraction decomposition).

Fix and why it works. Charge *re-selection* in refine (`best_charge_by_fit`) re-scores the candidate charges $\{z/2, z, 2z\}$ by the envelope-fit cosine, which *does* penalise the intervening peaks the $z=1$ comb leaves unexplained (they enter the cosine as $(t = 0, o = I)$ slots that inflate $\|o\|$ with no numerator contribution). The metric places the mono where the whole envelope is explained, not where a subset comb happens to land. This made detector-anchored shift decon + envelope-fit recharge the pipeline default and took recall $89.5\% \rightarrow 95.0\%$ (charge $85.3\% \rightarrow 91.0\%$); the “absent-miss” class fell $13 \rightarrow 3$. EFLDVIK now resolves at $862.488 \text{ Da} / z=2$.

Generalisable insight. A charge scorer used for cross- z NMS must reward *explaining the intervening teeth*, not just matched intensity; equivalently, it must penalise the signal a lower-charge subset comb ignores. RawSum’s un-normalised matched filter cannot do this by construction, which is the structural motivation for the envelope-fit cosine.

3.2 Charge-doubling (real $z=z$ labelled $z=2z$)

Symptom. The mirror image: in crowded windows, charge re-selection promotes a genuine $z=z$ feature to $z=2z$, doubling its mass and losing the peak. This is the danger the recharge machinery itself introduces, and it was widespread, not confined to the traced case.

Mechanism. Because the $z=z$ teeth are a subset of the $z=2z$ grid, and in a crowded window co-eluting *interferents* happen to fill the intervening $z=2z$ teeth, the $z=2z$ hypothesis “explains more” and wins a strict argmax over $\{z/2, z, 2z\}$. A relative margin alone cannot save the true charge here: the correct $z=z$ envelope may genuinely fit *poorly* (weak, overlapped), so there is no relative gap to defend it with.

Evidence. Traced on the sodium adduct **HVGDLGNVTAD[Na]K** (reference mass 1246.5918 Da , true $z=2$, RT 11.238 min , weak at 2.75×10^6 ; mono $m/z \ 624.309$). The detector had $z=2$ right, but strict-argmax recharge doubled it to $z=4$ (mass 2492.20). The measured fits are $z=1 \ 0.235$, $z=2 \ 0.258$, $z=4 \ 0.689$: the correct $z=2$ envelope *genuinely fits poorly*, and the spurious $z=4$ “win” is only a mediocre cosine (~ 0.69) earned by absorbing interferents on the denser grid. No margin over the true charge exists to protect it.

Fix and why it works. `best_charge_by_fit` keeps the detector’s own charge (`prefer_charge`) unless an alternative *both* beats it by `RECHARGE_PREFER_MARGIN = 0.05` *and* itself clears an *absolute* floor `RECHARGE_MIN_FIT = 0.85`. The absolute floor is the key discriminator: a spurious harmonic win is only ~ 0.69 , well under the floor, whereas a genuine re-charge (e.g. the light $z=2$ charge-halving recovery of Section 3.1) fits cleanly at ~ 0.98 and still fires. HVG now resolves at $1246.60 \text{ Da} / z=2$. Blocking spurious doubling took recall $96.6\% \rightarrow 97.1\%$ and charge-matched $92\% \rightarrow 94\%$ — a dataset-wide effect.

Generalisable insight. Harmonic ($z \leftrightarrow 2z$) ambiguity is intrinsic to isotope-comb matching and cannot be resolved by relative fit comparison alone. It requires (a) *loyalty* to a prior (here the detector’s own charge, itself derived from independent evidence) and (b) an *absolute* quality gate, so that a charge is only overridden by an alternative that is good in absolute terms, not merely better than a poor incumbent. Relative margins are necessary but not sufficient when the true hypothesis can itself fit badly.

4 Anchor selection

4.1 The distant grab

Symptom. A feature’s deconvolution places the monoisotope several isotopes away, on a different species, drifting the mass by multiple ^{13}C units.

Mechanism. When deconvolution anchors on the *tallest peak in a wide window*, and that window contains a stronger co-eluting *different* species several isotopes above the true feature, the anchor lands on the neighbour and the whole averagine template is placed relative to it — putting the mono on the interloper. This “distant grab” was the root of $\sim 78\%$ of the disagreements in a four-way (classic $\times$ shift $\times$ composite $\times$ apex) decon comparator.

Fix and why it works. Anchor the shift views on the *detector’s own* most-abundant claimed peak (`four_way_decon_detector_anchor`; the tallest of `feature.peaks`), so the deconvolution cannot grab a foreign peak at all. A neighbour-grid gating variant (`shift_decon_gated`, which excludes peaks on a co-eluting *detected* neighbour’s isotope grid unless they are on this feature’s own grid) was also implemented, but it only reduced disagreement to 60.8% because *undetected* neighbours slip through; detector-anchoring collapsed disagreement 64.2% $\rightarrow$ 45.4% (unanimous 35.8% $\rightarrow$ 54.6%) and won.

Averagine keyed on the most-intense mass, not the monoisotope. A related design choice underlies the anchor: the averagine template is keyed by the observed *most-intense* (tallest) peak’s mass, not by a monoisotope guess. The tallest peak is the highest-SNR, most accurately centroided, hardest-to-misassign point in the envelope; the monoisotope — especially for heavy peptides whose mode sits above it — is exactly the fragile quantity being recovered. Keying the template on the mono would be circular. The template’s mode tooth is pinned at the anchor and the mono falls out as $\text{anchor} - \text{mode} \cdot \Delta$, corrected by the shift search. This mirrors the classic deconvolution’s use of the observed most-intense peak as reference.

Generalisable insight. Anchor a deconvolution on the most robust observable, not on the quantity you are trying to estimate. When a window may contain foreign species, the safest anchor is one an upstream stage already committed to (the detector’s claimed peak), because it cannot drift onto a neighbour.

4.2 The single apex scan beats the averaged composite

A ground-truth cut (per-feature verdicts joined to the FlashLFQ peaks) showed that when the apex-composite and single-apex views disagree, the *single apex scan* is correct 70% of the time (classic decon) versus 16% for the composite; per-view accuracy is classic-apex 90.2%, shift-apex 87.7%, shift-composite 86.7%, classic-composite 83.9%. Averaging even a small window pulls in

co-eluting interference; the apex is a higher-SNR snapshot. This motivates the domain rule that the composite must stay small (Section 10) and the pipeline’s default to the shift-apex view. *Caveat:* these accuracies are from one dataset with narrow (~ 1.9 s FWHM) peaks; on longer gradients a slightly wider composite may be preferable.

5 Cross-charge resolution and knitting

The resolve stage knits refined features into peptide-level features. Its two sub-decisions — *which* features to merge, and *which* mass to adopt for the merged group — each have a characteristic failure.

5.1 Merging separately-eluting species as false off-by-ones

Symptom. A correctly-placed feature is dragged onto a wrong mass because it is knitted with a *different* co-eluting species ~ 1 Da away, which then out-votes it.

Mechanism. `features_link` bridged any two features within ± 2 ^{13}C in mass and within the apex-RT window, with *no co-elution check*. A different peptide ~ 1 Da away that merely drifted inside the 0.1-min window was knitted as a false off-by-one; being stronger, it dragged the consensus mass onto itself.

Evidence (must split). For **ECCHGDLLECADDRADLAK** ($z=4$, apex 12.153 min, narrow elution window [12.153, 12.169]), the corrected $z=4$ feature was wrongly merged with a *different* co-eluting $z=2/z=3$ species at ~ 2247.94 Da (apex 12.252 min, outside $z=4$ ’s window). The stronger interloper out-voted the corrected $z=4$ and the true mono was lost. Note additionally that at $z=3/z=2$ ECC’s true mono peak ($749.986 / 1124.475$ m/z) is *genuinely absent* from the centroids (its envelope runs +1, +2, +3, +4), so those charges legitimately fit +1; only $z=4$ sees the mono — see Section 6.

Evidence (must stay knitted — counter-example). **DSCQGDSGGPVVCNGQLQGIVSWGYYG-CAQK** (reference mass 3183.3808 Da, $z=3$, RT 15.414 min) genuinely co-elutes at $z=3/z=4/z=5$ (all apex 15.414 min). These are the *same* peptide at different charges and must remain knitted; the fix must not break them.

Fix and why it works. Off-by-one links ($k \neq 0$) are gated on tight co-elution (`features_coelute_tightly`): each feature’s apex must fall inside the other’s [start_rt, end_rt] window (padded by ± 0.6 s). Pure same-mass cross-charge links ($k = 0$, the same peptide at another charge) are untouched, so DSC’s genuinely co-eluting charges still knit, while ECC’s $z=4$ splits from the $z=2/z=3$ interloper whose apex lies outside $z=4$ ’s narrow window. This was the single largest lever of the session: +9 recall (part of $589 \rightarrow 598$), not just the traced peptide.

Generalisable insight. Chromatographic co-elution is a discriminator the mass domain alone does not have. Two species ~ 1 Da apart are indistinguishable to an off-by-one mass test, but their elution profiles are not: requiring mutual apex-in-window co-elution before treating a ~ 1 Da mass difference as an off-by-one uses the orthogonal RT axis to separate real mono-placement disagreements (same peptide) from coincidental mass neighbours (different peptides).

5.2 Consensus voting by count or intensity rather than fit

Symptom. The knitted group settles on the wrong monoisotopic mass even though a member placed it correctly.

Mechanism, two variants.

- (a) **Count.** Two off-by-one siblings out-vote one correct placement when the group votes by candidate count.
- (b) **Intensity.** A contaminated-but-stronger charge out-votes a clean but weaker one when ties are broken by summed intensity.

Evidence (intensity variant). **AAVTAFWGK** (reference mass 992.508 Da; FlashLFQ quantified it at $z=1$ although the precursor was $z=2$; RT 16.31 min; $z=1$ mono m/z 993.515). It is seen at $z=1$ (envelope contaminated by co-eluting interferents; refine mis-places the mono +1, envelope-fit 0.39) and at $z=2$ (clean; correct mono, fit 0.99). Both are lone singletons (support 1), so an intensity-only tiebreak took the stronger-but-wrong $z=1$ +1 mass and lost the true mono.

Fix and why it works. `resolve_mass_by_cross_charge` ranks mass clusters by the tuple (support, count, best of charge, fit). Distinct-charge *support* stays the *primary* key — a genuine multi-charge agreement must still out-rank a lone high-fit placement — but the envelope-fit is inserted as a tiebreak *before* intensity, so when support ties the clean placement wins. **AAVTAFWGK** now resolves at 992.517 Da / $z=1; z=2$. Recall 598 $\rightarrow$ 599. Critically, promoting summed fit to the *primary* key regressed recall (Section 8.2); fit works only as a tiebreak after support.

Generalisable insight — cross-charge redundancy is the central resource. When one charge state is ambiguous, contaminated, or missing its monoisotope, another charge of the same peptide often resolves it. But this redundancy pays off *only if resolution weights quality over quantity and intensity*. A stronger contaminated charge and two off-by-one siblings are both “more” evidence pointing the wrong way; fit quality is what distinguishes a clean placement from a loud wrong one. This is arguably the chapter’s central positive insight: exploit cross-charge redundancy, but weight it by fit, with independent multi-charge agreement (support) as the primary anchor so a single high-fit outlier cannot override a genuine consensus.

6 Data-limited and inherent-ambiguity cases

Not every off-by-one is an algorithm bug. Two classes are genuine data limitations, and honesty about them bounds what any algorithm can achieve.

Missing monoisotope at some charges. For some peptides the true monoisotope peak is simply *absent* from the centroids at certain charges — contamination or peak-picking drops it — while present at others. **ECC** again: at $z=2/z=3$ the mono m/z (1124.475 / 749.986) is not in the centroids at all (the observed envelope begins at +1), so the envelope legitimately starts one ^{13}C above the mono and the fit *correctly* prefers the +1 placement at those charges. The “error” is a real data gap. It is resolvable only by trusting the charge that *does* see the mono ($z=4$ here) — which is exactly why cross-charge resolution (Section 5.2) with quality weighting, not per-charge fit, is the right backstop.

Contaminated envelopes in crowded regions. Even the *correct* charge can fit poorly when the window is crowded: AAVTAFWGK $z=1$ fits 0.39; HVG $z=2$ fits 0.258. In such regions absolute cosine values are low across all hypotheses and relative fit comparisons become unreliable — the reason Section 3.2’s absolute floor and Section 5.2’s support-primary ranking are needed rather than pure argmax-of-fit.

Insight. Distinguish genuine algorithm errors from data-limited inherent ambiguity. Some “misses” are cases where the monoisotope is not in the data at a given charge, or where the reference itself may be questionable. Chasing these with ever-more-aggressive placement heuristics risks regressing the cases that *are* recoverable (see the negative results). The correct response is architectural: rely on the charge that sees the clean signal, via quality-weighted cross-charge consensus.

7 Detector-level artifacts

Greedy tallest-first claiming. The detector claims peaks tallest-first, and each peak is owned by exactly one feature (a claim partition). This is what makes cross- z NMS work (a claimed harmonic cannot re-fire), but it creates artifacts that interact with refinement: in dense regions a real isotope tooth can be claimed first by a marginally-taller *spurious* neighbour, so the claim partition is noisy. This directly explains why refine-stage peak censoring regressed recall (Section 8.4): deleting “already-claimed” peaks removes real teeth that the noisy partition mis-assigned.

The RawSum charge score. As formalised in Eq. 1 and Section 3.1, the default detector charge score is an un-normalised $\sum_k \sum_s w_k g_s I_{k,s}$ that neither penalises missing predicted teeth nor unexplained observed peaks. It is the upstream root of the charge-halving class and the structural motivation for the envelope-fit cosine downstream. A normalized matched-filter variant (`ScoreModel::NormalizedNoiseFloor`) was implemented and measured; it gave +1.4% recall but −0.8% charge and so failed its “charge up or flat” bar, and **RawSum** remains the detector default — the charge fix was moved downstream into refine’s envelope-fit recharge, which is where the cosine can see the whole envelope on a clean apex view.

Insight. A greedy claim partition is an implementation convenience, not ground truth; downstream stages must not treat “claimed by another feature” as “not part of my envelope.” And the natural place to fix a charge error is wherever the richest evidence lives — here, refine, where a high-SNR apex composite and a completeness-aware metric are both available.

8 Negative results

These experiments regressed recall and were reverted. They are recorded because each rules out a plausible direction and explains *why* the intuition failed.

8.1 Widening the envelope-fit window below the mono

To attack the window blind spot (Section 2.1) directly, the envelope-fit window’s lower reach was extended from $\text{mono} - 0.5 \cdot \Delta$ to $\text{mono} - 2.5 \cdot \Delta$ so the metric could see the peak below a too-high mono. This regressed recall both globally (to 586) and when scoped only to the walk-back (587). **Why it fails:** a wider lower window admits *legitimate* below-mono signal — the +1/+2 teeth of a genuine lower-mass co-eluting species, or noise — as spurious “unexplained” penalty against *correct*

placements, depressing good cosines more than it rescues bad ones. The targeted, self-protecting walk-back (which only *prefers* a lower mono when it fits at least as well) succeeds precisely because it does not globally change what counts as in-window.

8.2 Fit-weighted consensus as the primary ranking key

Ranking mass clusters by *summed decon_score* as the *primary* key (instead of distinct-charge support) regressed recall. **Why it fails:** two off-by-one siblings still out-*sum* one correct member, so summing fit reproduces the count failure it was meant to cure. This is the crucial contrast with Section 5.2’s working fix, where fit is only a *tiebreak after support*: independent multi-charge agreement must remain the primary anchor, or a pair of concordant-but-wrong placements overrides a lone correct one.

8.3 Over-broad walk-back gating

The mono walk-back was tried at all charges, with only a 2000 Da mass floor, and with an $\text{argmax}(\text{template}) \geq 1$ mode gate (~ 1500 Da). All regressed (to 587–588). **Why it fails:** at low charge / mid mass the fit metric’s window cannot see below the mono (the same blind spot), so it scores a +1 placement above the true one and the walk-back *over-corrects*, walking a correct mid-mass low-charge mono down to a wrong lower mass. The gate must key on the physical driver of the error — high mass *and* high charge, where the mode genuinely sits above the mono — hence ≥ 2200 Da and $|z| \geq 3$.

8.4 Refine-stage subtractive censoring

Removing peaks claimed by *other* features from a feature’s composite before deconvolution (to clean chimeric windows) regressed recall substantially: baseline 88.2% / 84.0% charge $\rightarrow$ grid-gated censor 84.7% / 77.9% $\rightarrow$ full censor 83.9% / 75.6%. **Why it fails, two mechanisms:** (1) classic deconvolution scores an envelope using the same neutral mass at *other* charge states (*observe_adjacent_charge_states*); those cross-charge peaks are off this feature’s m/z grid, so censoring deletes the very cross-charge evidence decon relies on — which is why the *charge*-recall column drops hardest (–6%). (2) The averaging normalises per-scan by TIC, so removing peaks reshuffles the composite. Combined with the noisy claim partition (Section 7), censoring deletes real teeth. **Lesson:** already-assigned peaks are not merely interferences — deconvolution exploits neighbouring and cross-charge peaks, so a claim partition must not be read as a segmentation.

8.5 Coverage depth as a recovery lever

Raising the coverage target to detect more features ($0.90 \rightarrow 0.95$; $\sim 96\,000 \rightarrow 143\,000$ features) recovered *zero* additional reference peaks and left the miss set byte-identical (Section 3.1). **Why it fails:** the misses were not undetected — they were detected and then mislabelled (charge, mono, or knitting). Detection depth is the wrong knob when the loss is downstream; more darts do not help when the darts already land and are then mis-scored.

9 Synthesis: design principles

The failures above, though they surface at different stages, teach a small set of transferable principles for isotope-envelope feature detectors.

1. **A scoring metric must see signal on both sides of its hypothesis.** A one-sided window (mono -0.5Δ upward) is structurally blind to an error in the excluded direction (mono-too-high). Either make the metric two-sided or add an explicit, tightly-gated re-scoring against the alternatives whose windows reach the missing evidence (Sections 2.1, 8.1).
2. **Harmonic ambiguity needs absolute quality gates, not just relative comparison.** The $z \leftrightarrow 2z$ degeneracy is inherent to comb matching. Because the true hypothesis can itself fit poorly in crowded windows, a relative margin cannot protect it; combine loyalty to an independent prior with an *absolute* fit floor so a charge is overridden only by an alternative that is good in absolute terms (Sections 3.1, 3.2).
3. **Chromatographic co-elution is a discriminator the mass domain lacks.** Species ~ 1 Da apart are indistinguishable to an off-by-one mass test but separable by elution profile. Gate mass-based knitting on mutual apex-in-window co-elution (Section 5.1).
4. **Exploit cross-charge redundancy, weighted by quality.** A peptide’s other charge states are the richest resource for resolving an ambiguous, contaminated, or mono-missing charge — but only if consensus weights fit quality, with independent multi-charge *support* as the primary key so neither a loud contaminated charge nor a pair of concordant-but-wrong siblings can override a clean placement (Sections 5.2, 8.2).
5. **Anchor estimation on the most robust observable.** Anchor deconvolution on the highest-SNR peak (or an upstream committed claim), never on the fragile quantity being estimated; this prevents both the distant grab and circular mono-keying (Section 4).
6. **Distinguish algorithm errors from data-limited inherent ambiguity.** Some misses are genuine data gaps (mono absent at a charge) or questionable references; the correct response is architectural (trust the charge that sees clean signal), not an ever-more-aggressive heuristic that regresses recoverable cases (Section 6).
7. **Adversarial and negative experiments are load-bearing.** Four of the most tempting “obvious” improvements (wider window, fit-primary consensus, broad walk-back, censoring) all regressed recall, each for an instructive reason. Measuring and reverting them — and recording *why* — is what prevents the same intuitions from being re-implemented (Section 8).
8. **Fix an error where the richest evidence lives.** The charge-halving error originates in the detector but is best fixed in refine, where a high-SNR apex composite and a completeness-aware metric are both available (Sections 3.1, 7).

10 Open problems and remaining misses

After the fixes above, ~ 18 reference peaks (602/620) remain unrecovered (an earlier snapshot: 22 misses = 3 genuinely absent +19 off-by-one). Off-by-one remains the single largest miss category. The open problems:

- **Contaminated single-charge features with no clean sibling.** The cross-charge backstop (Section 5.2) needs a second charge that sees the clean signal. When a peptide is observed at only one charge and that charge’s envelope is contaminated (low absolute fit, e.g. ~ 0.25 – 0.39), there is no clean vote to prefer and relative fit is unreliable. A chimera-robust single-envelope discriminator (or a genuine multi-envelope NNLS unmixing of the crowded window) is the plausible next step; a naive cosine corrector for this case regressed recall in earlier work.

- **Data-dependent averaging-window sizing.** The composite is hard-capped at 3 scans (apex ± 1), tuned to the CA/Lumos 10-min gradient’s narrow ~ 1.9 s FWHM. This is not appropriate for longer gradients, where a feature spans more MS1 scans and averaging only 3 would discard real signal and clip the elution. The constant should be replaced by a scan count derived from the measured chromatographic FWHM (the detector already measures FWHM via XIC half-max for its RT σ), so it adapts to gradient length instead of being hard-coded.
- **Genuine data gaps and reference quality.** Some remaining “misses” are cases where the monoisotope is absent from the centroids at every observed charge, or where the reference peak itself is questionable. These bound the achievable recall and should be characterised (via the miss-explorer envelopes) rather than optimised against, to avoid regressing recoverable cases.
- **Precision.** The shift-apex default roughly doubled the resolved-feature count versus the classic path (the classic **None**-drop was an implicit quality filter). Recall improved, but a precision/false-discovery accounting against the reference is an outstanding check.

Summary tables

Stage	Failure / mechanism	Example peptide	Fix
Refine (mono)	Heavy off-by-one-too-high; fit-window blind below mono	ECCHGDLLECADDRADDAK ($z=4$, 2246.94 Da)	Walk-back, gated mass ≥ 2200 Da & $ z \geq 3$
Detect / Refine (charge)	Charge-halving: even $z=2$ isotopes form a valid $z=1$ comb	EFLDVIK ($z=2$, 862.48 Da)	Envelope-fit recharge over $\{z/2, z, 2z\}$
Refine (charge)	Charge-doubling: $z=z$ teeth are a subset of $z=2z$; interferences fill gaps	HVGDLGNVTAD[Na]K ($z=2$, 1246.59 Da)	Detector-charge loyalty + absolute fit floor 0.85
Refine (anchor)	Distant grab: anchor on stronger foreign peak in wide window	(four-way comparator, dataset-wide)	Anchor on detector’s own claimed peak
Resolve (knot)	Separately-eluting species merged as false off-by-one	ECC $z=4$ vs $z=2/z=3$ inter-loper	Co-elution gate on off-by-one ($k \neq 0$) links
Resolve (vote)	Consensus by intensity/count overrides clean placement	AAVTAFWGK ($z=1$ fit 0.39 vs $z=2$ 0.99)	Fit as tiebreak <i>after</i> support (not primary)
Data-limited	Mono absent from centroids at some charges	ECC at $z=2/z=3$	Trust the charge that sees the mono ($z=4$)
Detect	RawSum ignores missing teeth & unexplained peaks	(root of charge-halving class)	Envelope-fit cosine downstream

Table 1: Failure-mode taxonomy: stage, root mechanism, traced example, and fix.

Step	Fix applied	Recall / 620	Charge-matched
Baseline (classic decon)	—	555 (89.5%)	529 (85.3%)
Detector-anchored shift-apex + recharge	Secs. 3.1, 4.1	589 (95.0%)	564 (91.9%)
Mono walk-back	Sec. 2.1	589	566
Co-elution knitting gate	Sec. 5.1	598 (96.5%)	570
Fit tiebreak in consensus	Sec. 5.2	599 (96.6%)	570–573
Anti-doubling fit floor	Sec. 3.2	602 (97.1%)	584 (94.2%)

Table 2: Recall progression across the session’s fixes on CA/Lumos cov90 vs FlashLFQ. Intermediate charge-matched values (566, 570, 573) are reported where the source commits recorded them; the 589 $\rightarrow$ 602 recall trajectory and 529/564/584 charge endpoints are the load-bearing figures. The 89.5% $\rightarrow$ 95.0% jump is the shift-apex+recharge default; the walk-back alone left *recall* flat at 589 while lifting charge to 566 (it corrects mass placement on already-counted features), with the co-elution gate delivering the next +9.

Sources and reproducibility

All numbers are drawn from the pipeline’s commit history and verified where possible against the data files. The reference peptide masses, charges, RTs, and m/z values in Sections 2–6 were re-read directly from `AllQuantifiedPeaks.tsv` (wide-tolerance search) and agree with the traced values: ECC $z=4$ 2246.9355 Da RT 12.153 m/z 562.997; HVG[Na] $z=2$ 1246.5918 Da RT 11.238 m/z 624.309 intensity 2.75×10^6 ; AAVTAFWGK $z=1$ 992.508 Da RT 16.31 m/z 993.525; DSC $z=3$ 3183.3808 Da RT 15.414 m/z 1062.478; EFLDVIK $z=2$ theoretical 862.480 Da RT 14.70 m/z 432.251. The full envelope-fit metric derivation (including the $S^2 = R^2$ explained-energy identity and the EFLDVIK completeness/explained-fraction table) is in the companion note *An Envelope-Fit Metric for Charge-State and Monoisotope Selection* (`agent_info/Envelope-Fit-Metric.tex`). Fit scores quoted for HVG ($z=1$ 0.235, $z=2$ 0.258, $z=4$ 0.689), AAVTAFWGK ($z=1$ 0.39, $z=2$ 0.99), and the walk-back (0.934 at +1) are from the traced-run diagnostics recorded in the commit messages and were not independently re-derived here.